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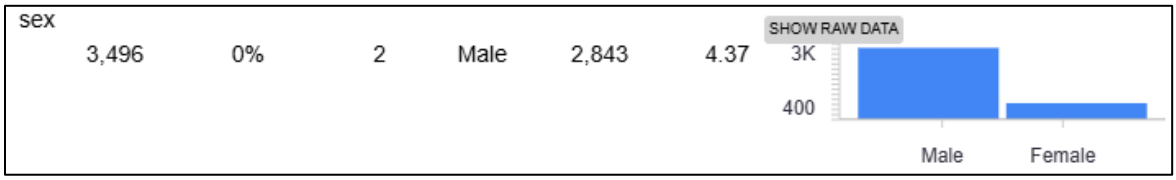
95-891 Intro to AI

Dr. David Steier

HW 5 – Model Fairness | COMPAS

(Q1)

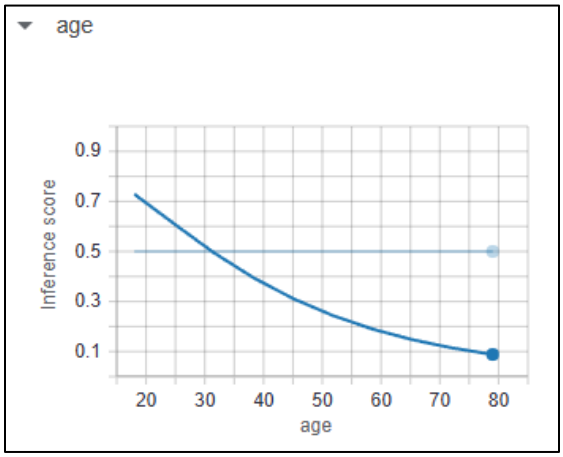


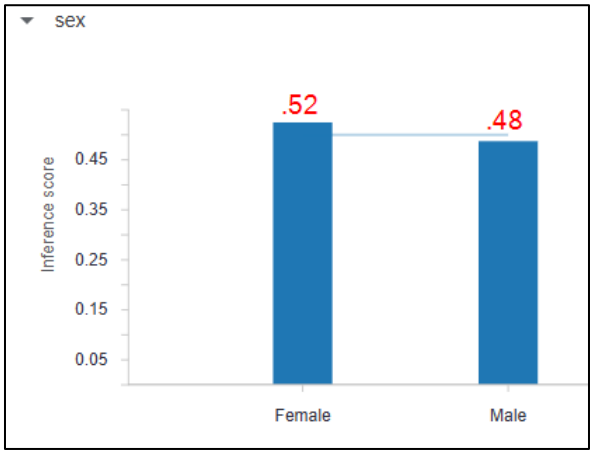
Males: 2,843 Females: 653

Because there are far fewer women in the data, any accuracy or fairness metrics computed for women are based on a much smaller sample, making them less statistically reliable.

This model, if training data is also as skewed as the WIT data, risks being poorly generalized for women and biased in its results.

(Q2)





The partial dependence plots for Age and Sex set out to visually measure the effect of these features on the predicted ratio of being high-risk to [criminally] reoffend. The range for age in this dataset is 18-79 and you can see an obvious downward trend in recidivism rate with age. Young people are considered high risk. With the steep negative and roughly linear relationship, age has a very strong influence on the COMPAS model's predictions. This likely reflects real-world patterns, however risks overly punishing younger offenders because of this.

The sex PDP shows minimal difference between male and female predictions, suggesting sex is not a strong direct driver of the model's output. However, from this singular plot we cannot conclude the absence of bias from the model. Sex could still interact with other features in ways not captured here. The dataset imbalance from Q1 means the female subgroup's performance may be underrepresented.

(Q3)

The accuracy for *recidivism_within_2_years* when grouped by age varies widely across age groups. ~56% for the youngest group (18-23) and ~89% for the oldest (74-79) with a steady trend upwards in accuracy. As we determine before, the model broadly flags young people as high risk, even those who won't reoffend. Accuracy is highest in the 23–38 age range, which also contains the most datapoints, suggesting the model is best calibrated for the age groups it saw most frequently during training. For older defendants, accuracy appears to recover in the raw numbers, but the sample sizes become extremely small (as few as 9 people in the 74–79 bracket). These figures may be statistically unreliable. The root cause of this variation ties directly to the PDP from Q2: because age is a strong predictor in the model, the model becomes overconfident at the high and low end of the data ranges. So, the model may cause increased false-positives in the young and false-negatives in the old.

(Q4)

The false positive and false negative rate of this model when slicing the dataset by gender only are roughly similar.

Group	Count	False Pos.	False Neg.
Male	2843	17.4%	17.0%
Female	653	15.8%	18.4%

Males have slightly higher false positives while females have slightly higher false negatives. Again, the female sample is much smaller so these metrics are less reliable.

When introducing race, the model’s poor performance becomes recognizable. There are two groups which show problems:

Male/African-American: Highest false positive rate at 24.8%. Model likely over-flags this group as high risk.

Male/Caucasian: Highest false negatives at 23.5%. Model under-flags white males.

I'm choosing to show groups with >50 individuals, as manually adjusting the thresholds for smaller groups may result in misrepresenting the recidivism rate for these groups if classification thresholds are taken to high or low extremes just to match the accuracy of larger groups.

Custom thresholds for 11 values of sex/race

Sort by Count

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)
Male/African-American	1604	0.63	15.6	22.2	62.2
Male/Caucasian	840	0.42	16.3	17.4	66.3
Female/African-American	300	0.58	17.7	15.3	67.0
Female/Caucasian	271	0.44	14.0	18.1	67.9
Male/Hispanic	263	0.37	14.4	19.0	66.5
Male/Other	121	0.3	11.6	24.0	64.5

These very slight adjustments to classification threshold for these gender/race combinations help to normalize the data from the default 0.5. These values align the false positive-negative rates without greatly affecting model accuracy scores for these groups. This reduces the score required to trigger a high-risk classification, thereby catching more of the true recidivism cases and reducing the damaging recidivism rates.

(Q5) Appendix attachments A, B, C

Demographic Parity: each group should receive positive predictions (flagged as high risk) at the *same rate*, regardless of actual recidivism rates. It ignores whether the predictions are correct.

Equal Opportunity: each group should have the same *false negative rate*, meaning people who *will* reoffend are caught at equal rates across groups. It focuses on not missing true positives equally.

Equal Accuracy: each group should have the same overall *accuracy*, balancing both false positives and false negatives together across groups.

	Demographic Parity	Equal Opportunity	Equal Accuracy
Male/African-American	0.68	0.65	0.52
Male/Caucasian	0.41	0.45	0.47
Female/African-American	0.59	0.61	0.70
Female/Caucasian	0.39	0.40	0.39
Male/Hispanic	0.32	0.34	0.76

Male/Other	0.23	0.24	0.23
Female/Hispanic	0.29	0.46	1.00
Female/Other	0.27	0.31	0.53
Male/Native American	0.99	0.99	0.99

Across the three fairness constraints, the required thresholds vary largely by sex/race combination. Under demographic parity, thresholds range from 0.23 (Male/Other) to 0.99 (Male/Native American). Equal opportunity produces a similarly wide range. Equal accuracy produces the widest spread, with Female/Hispanic requiring a threshold of 1.0 (a.k.a. never flagging anyone in that group) while Male/Other sits at 0.23.

This variation reflects the highlights from ProPublica: when base rates of recidivism differ substantially across groups, satisfying any single fairness requires very different decision thresholds per group; nearly an impossible task.

(Q6) Appendix attachments D, E, F

Cost ratio of 2 (FP weighted twice as heavily as FN)

	Demographic Parity	Equal Opportunity	Equal Accuracy
Male/African-American	0.89	0.84	0.75
Male/Caucasian	0.67	0.66	0.75
Female/African-American	0.81	0.77	0.86
Female/Caucasian	0.56	0.54	0.89
Male/Hispanic	0.59	0.64	0.28
Male/Other	0.33	0.32	0.49
Female/Hispanic	0.54	0.54	0.26
Female/Other	0.47	0.51	0.21

When FPs are weighted twice as heavily as FNs, the threshold is raised across all groups under every fairness constraint. This is because we are prioritizing the reduction of wrongful high-risk classifications at the cost of missing more true reoffenders. Under demographic parity, this produces the most extreme removal from baseline.

Male/African-Americans are assigned a threshold of 0.89, causing accuracy to drop to 53.2%. The model struggles to flag this group to achieve accurate prediction rates. Some groups require thresholds as low as 0.30 generating 30%+ false positive rates.

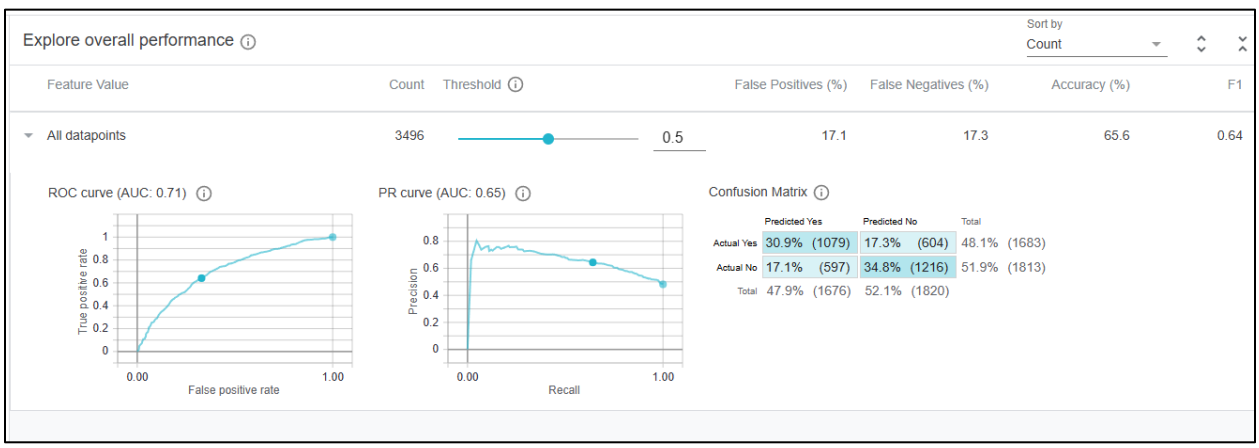
Equal opportunity emerges as the most suitable constraint under this asymmetric cost ratio. By equalizing the true positive rate across groups, it ensures that people who will genuinely reoffend have an equal chance of being correctly identified regardless of sex or race.

The accuracy of equal opportunity (57–76%) is more suited for this dataset when false positives are penalized using cost ratio. and avoids the demographic parity problem of rendering entire subgroups effectively invisible to the classifier.

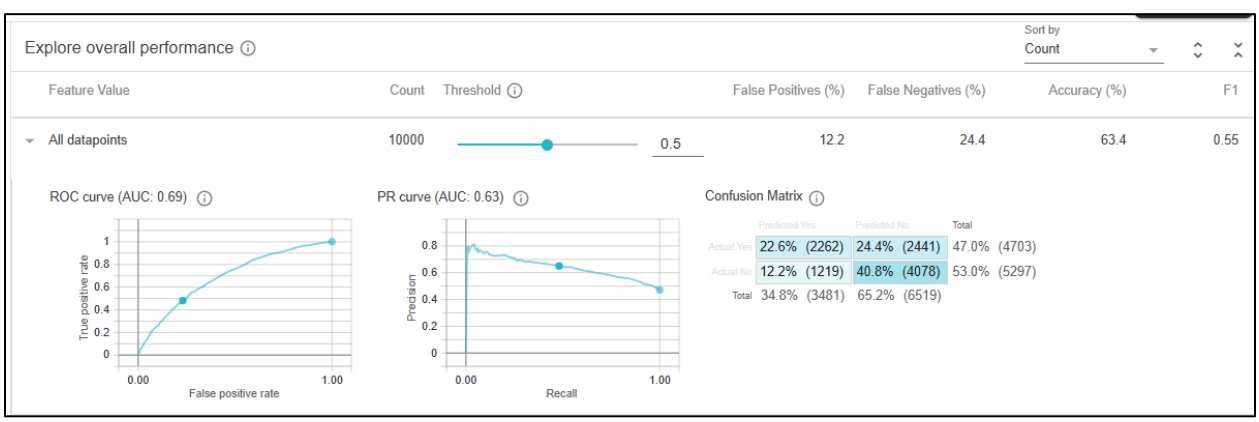
(Q7)

Removing 'sex' and 'race' inputs from the dataset greatly reduces the accuracy. We can see in the scatterplot that points have moved greatly to the high and low extremes of the graph. This indicates the model is having a very hard time with judging offenders on the remaining 5 models and is making wildly broad decisions.

With 'sex' and 'race' inputs: Accuracy: 65.6% FP: 17% FN: 17%



Without 'sex' and 'race' inputs: Accuracy: 63.4% FP: 12.2% FN: 24%



While removing these inputs does not seem to have has a major effect on the model's overall accuracy, it does have quite an impact on the false-positive and false-negative rate though. While accuracy is not greatly impacted, the FP/FN rate is not to be ignored.

Looking through the pre-sanitized dataset for columns that may be useful in increasing model accuracy, there are two columns not included in the original model that may prove useful in increasing the accuracy and robustness of the model:

[*decile_score.1*, *v_decile_score*]

General decile score is reoffense rate and *v_decile_score is* classification of violent recidivism into low, medium, and high-risk groups, generally:

- A decile score from 1 to 4 is considered low risk
- A decile score from 5 to 7 is considered medium risk
- A decile score from 8 to 10 is considered high risk

```
# Set column to predict
label_column = 'COMPASS_determination'

# Get list of all columns from the dataset we will use for model input or output.
input_features = ['v_decile_score', 'sex', 'race', 'age', 'priors_count', 'juv_fel_count', 'juv_misd_count', 'juv_other_count']
features_and_labels = input_features + [label_column]

features_for_file = input_features + ['recidivism_within_2_years', 'COMPASS_determination']
features_for_file
```

Adding the v_decile_score (violent recidivism decile score) or general decile_score as a feature will generally increase a model's accuracy, as these scores are already strongly correlated with the likelihood of recidivism. The results from model training shown below confirm this.

Regular 7-feature model accuracy:

```
Epoch 1/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 18s 60ms/step - accuracy: 0.5355 - loss: 5.6785
Epoch 2/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 0s 35us/step - accuracy: 0.5714 - loss: 2.0732
Epoch 3/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 12s 44ms/step - accuracy: 0.4673 - loss: 1.4075
Epoch 4/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 0s 34us/step - accuracy: 0.4286 - loss: 0.7039
Epoch 5/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 12s 44ms/step - accuracy: 0.6068 - loss: 0.6420
Epoch 6/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 0s 31us/step - accuracy: 0.7143 - loss: 0.6053
Epoch 7/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 24s 89ms/step - accuracy: 0.6902 - loss: 0.5817
89/89 ━━━━━━━━━━━━━━━━━━━━ 4s 41ms/step - accuracy: 0.7084 - loss: 0.5633
Model training completed!
```

v_decile_score added:

```
Epoch 1/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 19s 65ms/step - accuracy: 0.7142 - loss: 0.7815
Epoch 2/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 0s 36us/step - accuracy: 0.8571 - loss: 0.2514
Epoch 3/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 13s 46ms/step - accuracy: 0.8510 - loss: 0.3627
Epoch 4/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 0s 32us/step - accuracy: 1.0000 - loss: 0.0998
Epoch 5/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 13s 46ms/step - accuracy: 0.8679 - loss: 0.3218
Epoch 6/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 0s 31us/step - accuracy: 0.7143 - loss: 0.3778
Epoch 7/7
273/273 ━━━━━━━━━━━━━━━━━━━━ 13s 47ms/step - accuracy: 0.8820 - loss: 0.2902
89/89 ━━━━━━━━━━━━━━━━━━━━ 4s 46ms/step - accuracy: 0.8866 - loss: 0.2703
Model training completed!
```

Appendix:

Generative AI Declaration: "I did not use generative AI in producing this submission."

A Sex/Race – Demographic Parity (1 Cost Ratio FP/FN)

Demographic parity thresholds for 11 values of sex/race

Sort by Count

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)	F1
Male/African-American	1604	0.68	14.2	23.9	61.8	0.62
Male/Caucasian	840	0.41	16.8	17.1	66.1	0.61
Female/African-American	300	0.59	17.0	17.3	65.7	0.62
Female/Caucasian	271	0.39	17.3	12.9	69.7	0.64
Male/Hispanic	263	0.32	19.8	14.1	66.2	0.59
Male/Other	121	0.23	19.0	13.2	67.8	0.64
Female/Hispanic	42	0.29	23.8	9.5	66.7	0.56
Female/Other	36	0.27	19.4	13.9	66.7	0.57
Male/Asian	11	0.4	18.2	0.0	81.8	0.80
Female/Native American	4	0.55	0.0	0.0	100.0	1.00
Male/Native American	4	0.99	0.0	25.0	75.0	0.86

B Sex/Race – Equal Opportunity (1 Cost Ratio FP/FN)

Equal opportunity thresholds for 11 values of sex/race

Sort by Count

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)	F1
Male/African-American	1604	0.65	15.0	22.8	62.2	0.63
Male/Caucasian	840	0.45	13.3	18.7	68.0	0.61
Female/African-American	300	0.61	15.3	19.0	65.7	0.61
Female/Caucasian	271	0.4	16.6	16.6	66.8	0.58
Male/Hispanic	263	0.34	17.5	14.4	68.1	0.60
Male/Other	121	0.24	18.2	19.0	62.8	0.55
Female/Hispanic	42	0.46	4.8	14.3	81.0	0.64
Female/Other	36	0.31	11.1	13.9	75.0	0.64
Male/Asian	11	0.63	0.0	27.3	72.7	0.40
Female/Native American	4	0.55	0.0	0.0	100.0	1.00
Male/Native American	4	0.99	0.0	25.0	75.0	0.86

C Sex/Race – Equal Accuracy (1 Cost Ratio FP/FN)

Equal accuracy thresholds for 11 values of sex/race

Sort by Count

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)	F1
Male/African-American	1604	0.52	23.3	12.2	64.6	0.71
Male/Caucasian	840	0.47	10.8	20.5	68.7	0.60
Female/African-American	300	0.7	9.3	23.3	67.3	0.58
Female/Caucasian	271	0.39	17.3	12.9	69.7	0.64
Male/Hispanic	263	0.76	0.8	29.7	69.6	0.37
Male/Other	121	0.23	19.0	13.2	67.8	0.64
Female/Hispanic	42	1	0.0	31.0	69.0	0.00
Female/Other	36	0.53	0.0	30.6	69.4	0.27
Male/Asian	11	0.63	0.0	27.3	72.7	0.40
Female/Native American	4	0.38	25.0	0.0	75.0	0.86
Male/Native American	4	0.99	0.0	25.0	75.0	0.86

D Sex/Race – Demographic Parity (2 Cost Ratio FP/FN)

Demographic parity thresholds for 11 values of sex/race

Sort by: Count

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)	F1
Male/African-American	1604	0.89	4.4	42.4	53.2	0.35
Male/Caucasian	840	0.67	3.6	31.2	65.2	0.42
Female/African-American	300	0.81	3.7	33.0	63.3	0.41
Female/Caucasian	271	0.56	5.9	29.2	64.9	0.38
Male/Hispanic	263	0.59	3.4	25.1	71.5	0.48
Male/Other	121	0.33	6.6	31.4	62.0	0.34
Female/Hispanic	42	0.54	4.8	19.0	76.2	0.50
Female/Other	36	0.47	2.8	25.0	72.2	0.44
Male/Asian	11	0.47	9.1	27.3	63.6	0.33
Female/Native American	4	1	0.0	75.0	25.0	0.00
Male/Native American	4	1	0.0	100.0	0.0	0.00

E Sex/Race – Equal Opportunity (2 Cost Ratio FP/FN)

Equal opportunity thresholds for 11 values of sex/race

Sort by: Count

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)	F1
Male/African-American	1604	0.84	5.7	36.9	57.4	0.46
Male/Caucasian	840	0.66	3.6	29.9	66.5	0.45
Female/African-American	300	0.77	6.3	31.3	62.3	0.43
Female/Caucasian	271	0.54	7.4	27.3	65.3	0.42
Male/Hispanic	263	0.64	2.3	26.2	71.5	0.46
Male/Other	121	0.32	8.3	28.9	62.8	0.40
Female/Hispanic	42	0.54	4.8	19.0	76.2	0.50
Female/Other	36	0.51	0.0	25.0	75.0	0.47
Male/Asian	11	0.63	0.0	27.3	72.7	0.40
Female/Native American	4	1	0.0	75.0	25.0	0.00
Male/Native American	4	1	0.0	100.0	0.0	0.00

F Sex/Race – Equal Accuracy (2 Cost Ratio FP/FN)

Equal accuracy thresholds for 11 values of sex/race

Sort by: Count

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)	F1
Male/African-American	1604	0.75	9.8	29.8	60.4	0.56
Male/Caucasian	840	0.75	1.7	37.7	60.6	0.24
Female/African-American	300	0.86	1.3	38.3	60.3	0.27
Female/Caucasian	271	0.89	0.0	39.5	60.5	0.02
Male/Hispanic	263	0.28	27.4	11.8	60.8	0.58
Male/Other	121	0.49	4.1	35.5	60.3	0.23
Female/Hispanic	42	0.26	33.3	7.1	59.5	0.54
Female/Other	36	0.21	30.6	8.3	61.1	0.59
Male/Asian	11	1	0.0	36.4	63.6	0.00
Female/Native American	4	0.38	25.0	0.0	75.0	0.86
Male/Native American	4	0.99	0.0	25.0	75.0	0.86